

STATEMENT OF PURPOSE

I am **Dr. Monex Sharma**, currently working as an **Assistant Professor at Vignan University**, where I teach **B.Tech students in Robotics and Automation under the Department of Mechanical Engineering**. I hold a **PhD** with research experience in mobile robotics, autonomous navigation, and coverage path planning. My academic and professional experience includes doctoral research, industrial R&D, and teaching, which has developed my interest in pursuing a long-term academic career combining teaching and research.

Before entering academia, I worked as an **Associate Design Engineer at WCB Robotics India Pvt. Ltd.**, where I gained practical experience in robotics R&D and real-world robotic system development. My work involved areas such as robotic mechanisms, sensors, localization, sensor fusion, motion planning, control, kinematics, and system integration. This experience helped me understand the practical challenges associated with developing robotic systems and strengthened my interest in research and autonomous robotics.

My doctoral research focused on **“Terrain-Adaptive Multi-Objective Coverage Path Planning for Mobile Robots Using an Improved NSGA-II Algorithm.”** The research addressed coverage path planning for mobile robots operating in both flat and uneven terrains, with emphasis on generating efficient and collision-free paths. My research involved multi-objective optimization, path planning, robot motion, localization, sensor integration, simulation, and experimental validation. This work resulted in peer-reviewed journal publications and provided me with experience in identifying research problems, reviewing literature, developing algorithms, conducting experiments, analyzing results, and communicating research outcomes.

In my current academic role at **Vignan University**, I teach B.Tech students in **Robotics and Automation**. My teaching interests include Robotics, Mobile Robotics, ROS, Robot Programming, Control Systems, Computer Vision, Autonomous Navigation, and Artificial Intelligence for Robotics. I believe students learn robotics more effectively when theoretical concepts are supported by programming, simulation, laboratory experiments, and practical projects. I am therefore interested in providing students with opportunities to implement concepts through hands-on robotic applications.

My future research interests are focused on **autonomous mobile robots operating in uncertain and dynamic environments**. I am particularly interested in uncertainty-aware localization and navigation, sensor fusion, coverage path planning in environments with moving obstacles, and multi-objective optimization for autonomous robots. I also intend to explore applications such as autonomous search and rescue, inspection, and service robotics. My research approach will combine algorithm development with simulation and experimental validation using physical robotic platforms.

As a faculty member, I aim to involve students in research and development projects and provide them with practical exposure to robotics. I would like to develop robotic platforms that can be used as teaching and research testbeds for sensing, localization, navigation, ROS, design thinking, control system, and autonomous decision-making. I also intend to pursue externally funded research projects, publish quality research work, and develop collaborations with academic and industrial organizations. I am interested in joining **SRM University, Andhra Pradesh** and contributing to the **Department of Mechanical Engineering** through teaching, research, student mentoring, and robotics-related academic activities. My experience in industrial robotics R&D, doctoral research, and teaching B.Tech Robotics and Automation students has prepared me to contribute to both academic and research activities.

My objective is to build a sustained academic career focused on **effective teaching, meaningful research, and practical robotics education**. I look forward to contributing my experience and continuing to develop as an educator and researcher in robotics and autonomous systems at SRM University, Andhra Pradesh.

Dr. Monex Sharma

PhD

Assistant Professor, Vignan University